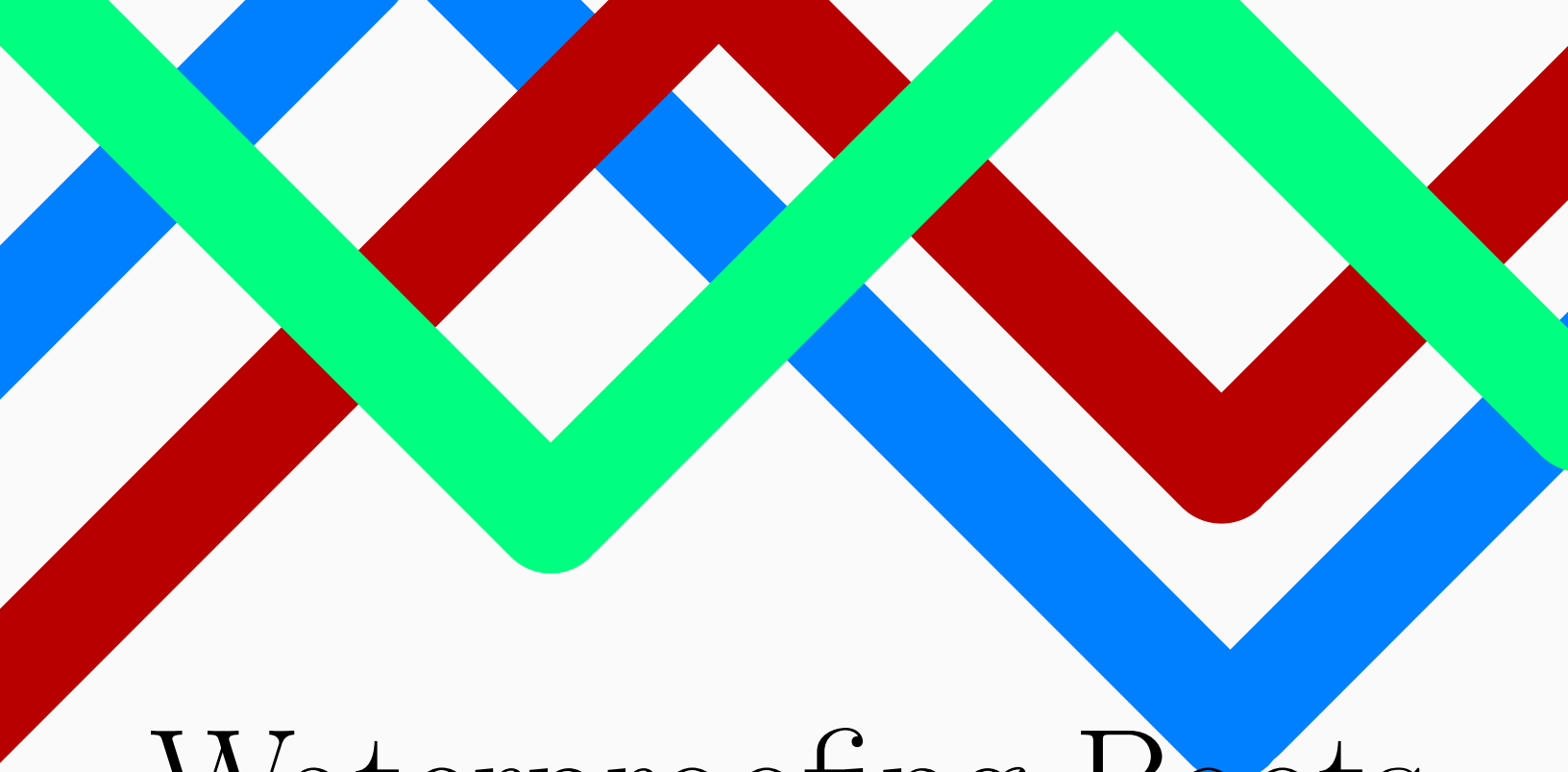




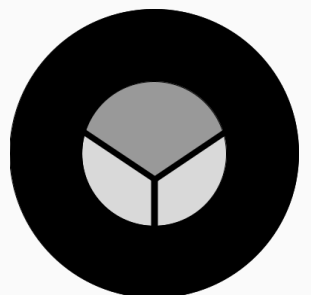
Waterproofing Boots

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DE Intro to Statistics I

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“Truth is ever to be found in the simplicity, and not in the multiplicity and confusion of things” - Issac Newton



1 Waterproofing Boots

“When in the Course of human events, it becomes necessary for one people to dissolve the political bands which have connected them with another, and to assume among the powers of the earth, the separate and equal station to which the Laws of Nature and of Nature's God entitle them, a decent respect to the opinions of mankind requires that they should dismantle the causes which impel them to the separation.

We hold these truths to be self-evident, that all men are created equal, that they are endowed by their Creator with certain unalienable Rights, that among these are Life, Liberty and the pursuit of Happiness.”

1.0.1 Manufacturing Questions

A manufacturer of boots plans to conduct an experiment to compare a new method of waterproofing to the current method. The appearance of the boots is not changed by either method. One hundred people who live in Seattle, where it rains frequently, volunteer to participate in the study. They will wear the boots as they normally would for 6 months. At the end of the 6 months, the boots will be returned to the company to be evaluated for water damage.

(a) Identify the following:

1) The explanatory variable.

- Old versus. New Waterproofing Method

2) The response variable.

- Difference in Water Damage

3) The experimental units.

- Subjects: People who live in Seattle participating in the study

(b) Identify some possibly confounding variables.

- Possible confounding variables include:

- Different human weights
- Different levels of rainfall
- Time spent in the rain
- Tragedy of the Commons may lead to people abusing the free shoe.

(c) Describe a design for this experiment that uses the 100 volunteers. Include commentary on how it would be implemented. Recall, there are three experimental designs in the Great Bay Stat curriculum:

- Completely randomized design
 - To implement “Completely Randomized Design,” I would assign a unique identification to each volunteer. Then, I would randomly choose 50 volunteers to wear the experimental boot, and other 50 volunteers to wear the control boot. This way, each person would have the ability to wear the boot for 6 months, and they would send it back to evaluate for water damage. Each volunteer would be instructed to wear the boot as if they would a regular boot.
- Randomized block design

- To ensure that there are homogenous groups (blocks) selected with this experiment, I would make groups of the 100 volunteers based on age-groups. I would split the volunteers into Under 20, 20-65, and 65+. This way, each homogenous group will have different times wearing the shoe; volunteers under 20 and people ages 65+ are likely to spend a lot of time outdoors, while people between the ages of 20-65 are likely to be preoccupied with their job. Then, I would split the group in half; 50% would wear the experimental group, and the rest would wear the control group. This way, each person would have the ability to wear the boot for 6 months, and they would send it back to evaluate for water damage. Each volunteer would be instructed to wear the boot as if they would a regular boot.
- Matched pairs design
 - In this case, I could match volunteers most closely based on age, gender, and weight. This would then allow me to assign one of the two with the experimental boot, and the other with the control group. This way, each person would have the ability to wear the boot for 6 months, and they would send it back to evaluate for water damage. Each volunteer would be instructed to wear the boot as if they would a regular boot.

I like the "Randomized Block Design" most in this case due to its ability to split up the volunteer group into homogenous subgroups.

- (d) Three of the four principles of experimental design are necessary for any well-designed experiment: control, randomization, replication. For the design that you chose in part charlie, describe the why and the how you would implement each of the three principles.
- I chose "Randomized Block Design" because it exemplifies three out of the four principles of experimental design. The "RBD" experiment has a control group; half of the participants wear a regular boot to express causation in the results. Randomization is also key; this experiment is randomized each time such that each combination has an equal chance of occurring. Furthermore, this experiment is also able to be replicated with relatively the same results.
- (e) Could your design be double blinded? Explain.
- Yes; I would just not tell the person if the boot is the control group or the experimental group, removing the placebo bias. People do not if they are receiving the experimental group or the control group, so they are likely to wear the boot relatively the same.
- (f) Could you generalize the results of this study to customers who purchase these boots from across the U.S.? Explain.
- No; these are 100 participants from Seattle. Seattle has a particular climate; and the experimental design yields results specific to Seattle. However, if I were to expand this design to various places around the United States and yield positive results, I could generalize this to the US.